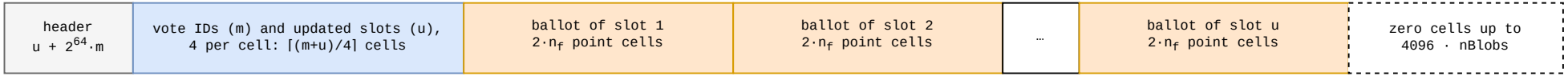
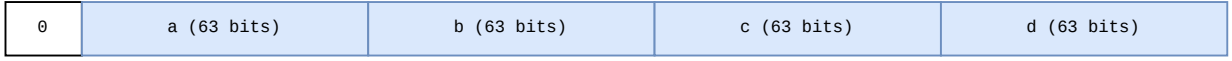
cells (32-byte big-endian elements of the BLS12-381 scalar field), in order:



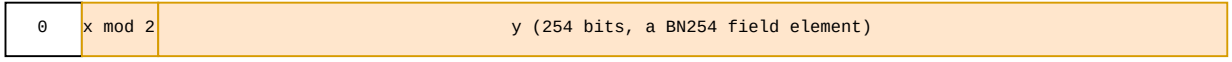
split into blobs of 4096 cells; $T = 1 + \lceil (m+u)/4 \rceil + 2 \cdot n_f \cdot u$ cells in total; $nBlobs = \lceil T / 4096 \rceil \le 32$

packed cell: $pack4(a, b, c, d) = a \cdot 2^{189} + b \cdot 2^{126} + c \cdot 2^{63} + d$



vote IDs enter without their top bit (all lie in $[2^{63}, 2^{64})$); slots are already below 2^{63}

point cell: $(x, y) \to y + 2^{254} \cdot (x \bmod 2)$



the identity (0, 1) encodes as 1; $p_{BN254} + 2^{254} < q_{BLS}$, so every cell is a valid field element

binding to the commitments

for each blob b with KZG commitment com_b:

$z_b = H(processID \,||\, stateRoot^{old} \,||\, com_b) \bmod q_{BLS}$

$y_b = p_b(z_b)$ (computed in the program)

$blobsDigest = H(com_1 \,||\, y_1 \,||\, \ldots \,||\, com_{nBlobs} \,||\, y_{nBlobs})$

the registry opens each (com_b, z_b, y_b) with the point-evaluation precompile against the versioned hash of the published blob

Slots are listed once, in increasing order, whether they hold a new vote, an overwrite, or a silent refresh. The results accumulator is not published; observers recompute it from the old and new ballots of the listed slots.